

Geography Ch-6 (Advanced) — Himalayan Glaciers, Snowline & GLOF

Advanced / India-applied | GS-I & GS-III (Disaster) | Source: D.R. Khullar + Majid Husain
Himalayan glaciers, snowline variation, Pleistocene glaciation evidence
Extra CA: South Lhonak Lake / Teesta GLOF (distinct from Must-Do's Gangotri retreat)

Facts from official sources | Inferences clearly marked | No fabricated data

HIGH

ADVANCED — Himalayan Glaciers & the Snowline (Khullar / Majid)

GS-I · India
Geography →
Glaciers (Source:
D.R. Khullar,
Majid Husain)

■ NEWS TRIGGER

The Himalaya holds ~15,000 glaciers — India's frozen water tower — with a snowline that swings widely with latitude and moisture, and which is now rising as the range warms.

■ INTRO & ORIGIN

Khullar records ~15,000 Himalayan glaciers between the eastern and western syntaxial bends. The **snowline** (lowest limit of perpetual snow) is not fixed — it depends on latitude, precipitation and topography.

*Most large Indian glaciers lie in the **Greater Himalaya and Trans-Himalaya (Karakoram, Ladakh, Zaskar)**; ancient moraines below present snouts prove a **Pleistocene glacial age** once reached far lower.*

■ KEY DATA

Region	Approx. snowline altitude	Note
Kashmir Himalaya	glaciers descend to ~2,500 m	Highest precipitation, lowest snout
Kumaon / Lahul	glaciers to ~3,660 m	Central Himalaya
Kumaun snowline	~5,200 m	Snowline (not snout)
Karakoram	~5,500 m	Largest glaciers (Siachen, Biafo, Baltoro, Hispar)
Tibetan side	~900 m higher	Desiccation / moisture scarcity

■ STATIC THEORY

- Khullar: about **15,000 glaciers** in the Himalaya, between the two syntaxial bends (east & west).
- **Snowline** = lowest level of perpetual snow; it varies with **latitude, precipitation and local topography**.
- Direct moisture rule (Majid): **more atmospheric moisture -> lower snowline** (and vice versa) — so the drier Tibetan side has a snowline ~900 m higher.
- Main glaciers lie in the **Greater and Trans-Himalaya (Karakoram, Ladakh, Zaskar)**; the Lesser Himalaya has only small glaciers (traces in Pir-Panjal, Dhauladhar).
- **Siachen, Biafo, Baltoro, Hispar** (Karakoram) are among India's/the region's largest glaciers; Pleistocene moraines prove past glaciation to far lower levels.

■ MUST-KNOW FACTS

1. ~15,000 Himalayan glaciers (Khullar), between the two syntaxial bends.
2. Snowline controls: latitude, precipitation, topography.
3. More moisture -> lower snowline (Tibetan side ~900 m higher).
4. Largest glaciers: Karakoram — Siachen, Biafo, Baltoro, Hispar.
5. Kashmir glaciers descend to ~2,500 m (wettest, lowest snout).

■ UPSC TRAPS

- ✗ The snowline is the same altitude across the Himalaya.
- ✓ It varies widely — lower in wet Kashmir (~2,500 m snout), higher on the dry Tibetan side (~900 m higher).
- ✗ Snowline and glacier snout are the same thing.
- ✓ Snowline = limit of perpetual snow; snout = the glacier's lower terminus (they differ).
- ✗ Most Indian glaciers are in the Lesser Himalaya.

✓ Most large glaciers are in the GREATER and TRANS-Himalaya (Karakoram etc.).

■ MAINS ANGLE

Explain why the Himalayan snowline varies with moisture and latitude, and assess the strategic and water-security importance of India's Karakoram glaciers.

■ STUDY LINK

Geography → India → Glaciers & Snowline
(Khullar: Glaciers of the Himalaya; Majid: Physiography)

■ MEDIUM UPSC RELEVANCE

MEDIUM

CA ANCHOR — South Lhonak Lake & the Teesta GLOF (Sikkim)

GS-I / GS-III ·
Glaciation applied
→ Glacial-lake
hazards

■ NEWS TRIGGER

In October 2023 a Glacial Lake Outburst Flood (GLOF) from South Lhonak Lake in North Sikkim surged down the Teesta, destroying infrastructure and damaging the Teesta-III dam.

■ INTRO & ORIGIN

GLOFs are the disaster face of glacial deposition: a **moraine-dammed glacial lake** that swells with meltwater can breach suddenly, releasing a catastrophic flood.

■ STATIC THEORY

- South Lhonak is a **moraine-dammed glacial lake** that grew as its feeder glacier retreated — area rose from ~0.42 sq km (2009) to ~1.48 sq km (2023).
- Trigger: heavy rain/cloudburst breached the moraine dam -> sudden outburst down the **Teesta**, damaging the Teesta-III hydropower project.
- Mitigation: glacial-lake monitoring, early-warning systems, and restricting development in the flood path (NDMA GLOF guidelines).

■ MUST-KNOW FACTS

1. GLOF = sudden breach of a moraine-dammed glacial lake.
2. South Lhonak (N Sikkim): ~0.42 sq km (2009) -> ~1.48 sq km (2023).
3. Oct 2023 outburst hit the Teesta & Teesta-III dam.

■ UPSC TRAPS

- ✗ GLOFs are caused only by earthquakes.
- ✓ They result from a moraine-dam breach — often triggered by rain/cloudburst or lake overtopping, not necessarily quakes.

■ MAINS ANGLE

Link glacial deposition (moraine dams) to GLOF risk and the case for glacial-lake early warning.

■ STUDY LINK

Geography → Glaciation → Glacial-lake Hazards / GLOF (applied CA)